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Graphical Abstract

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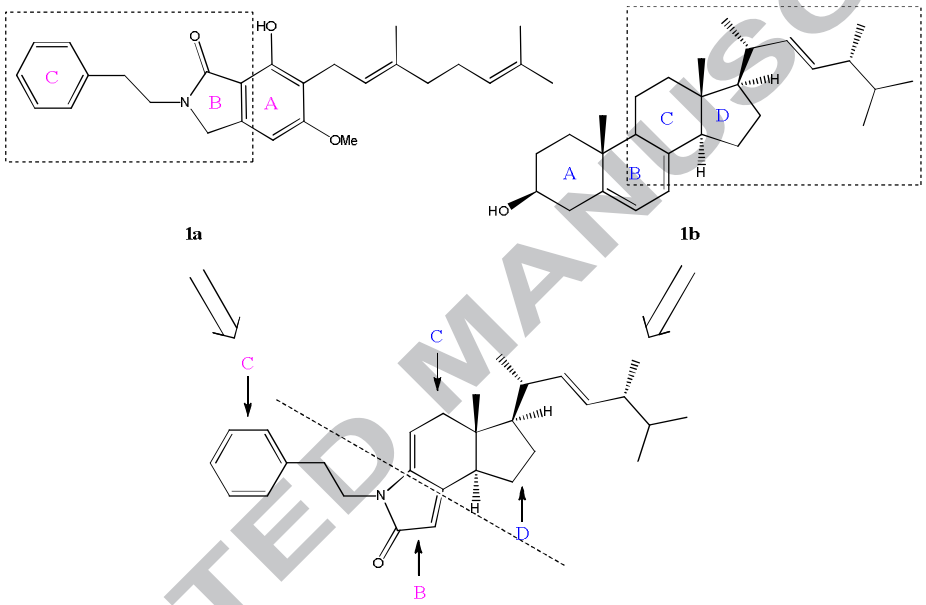

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Hericirine, a novel anti-inflammatory alkaloid from *Hericium erinaceum*

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ABSTRACT

A novel ergosterol conjunction-type alkaloid, hericirine (**1**), was isolated from the dried fruiting bodies of *Hericium erinaceum*. The overall framework of **1** was isolated, and combined with hericerin and ergosterol. Its structure was elucidated based on spectroscopic analysis using 1D and 2D NMR, IR, and HRESIMS techniques. We found that hericirine significantly inhibited protein expression of iNOS and COX-2 and reduced NO, PGE_2, TNF-α, IL-6 and IL-1β production in RAW264.7 cells exposed to LPS.

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Hericium erinaceum is an edible mushroom fungus belonging to the family Hericiaceae. The fruiting body of *H. erinaceum* is a common raw material of food dishes in Eastern Asia. Recently, this mushroom has been used as a food materials and is recognised as an important source of biologically active compounds with medicinal value.[1] In China, its fruiting bodies have been used for the treatment of dyspepsia, gastric ulcers, and enervation.[2] Several pharmacological studies on *H. erinaceum* have isolated compounds with various biological activities. Hericenone A causes cytotoxicity in HeLa cells;[3] hericenones C-E stimulates the synthesis of nerve growth factor;[4] and isohericerin and isohericenone reduce the viability of A549, SK-OV-3, SK-MEL-2, and HCT-15 cell lines.[5] Moreover, polysaccharides from *H. erinaceum* showed significant antitumour activity.[6] The present phytochemical and pharmacological study on *H. erinaceum*, searched for anti-inflammatory constituents and unique skeletons, which resulted in the isolation of a novel ergosterol conjunction-type alkaloid, hericirine (**1**) (Fig. 1). We report structural elucidation of **1** and its inhibitory activity against LPS-induced pro-inflammatory mediators and proteins in RAW264.7 cells.

Dried fruiting bodies of *H. erinaceum* were purchased from an herbal market, Geumsan, Chungnam Province, Korea in August 2013. The scientific name was identified by one of the authors (Young Ho Kim). A voucher specimen (CNU 13110) was deposited at the Herbarium College of Pharmacy, Chungnam National University. Dried fruiting bodies (2.5 kg) were extracted three times with MeOH (5 L) under reflux. The MeOH extract (320.0 g) was suspended in water and partitioned with $CHCl_3$, yielding $CHCl_3$ (90.0 g) and water (220.0 g) fractions. The $CHCl_3$ fraction (90.0 g) was subjected to silica gel (5.0 × 30 cm) column chromatography with a gradient of *n*-hexane-EtOAc-MeOH (25:1:0, 9:1:0, 5:1:0, 2.5:1:0, 1:1:0.1, 1:1:0.3, 0.5:1:0.5; 4 L for each step) to yield 8 fractions (Fr. 1A–1H). Fraction 1C was separated using silica gel (2.5 × 80 cm) column chromatography with a gradient of *n*-hexane-EtOAc (20:1 to 10:1, 10 L) to yield 11 sub-fractions (Fr. 1C-1–1C-11). Fraction 1C-2 was subjected to YMC (1.5 × 80 cm) column chromatography with MeOH-acetone-H_2O (5:5:1, 6.5:6.5:1, 8:8:1, 10:10:1; 1.0 L for each step) elution solvent to yield three sub-fractions (Fr. 1C-2-1–1C-2-3). Compound **1** (10.0 mg) was isolated from fraction 1C-2-2 using a silica gel column (1.0 × 80 cm) with *n*-hexane-EtOAc (10:1, 650 mL). Compound **1a** (1.4 g) and **1b** (1.8 g) were isolated from fractions 1D and 1G by recrystallization, respectively.

Figure 1. Structure of compound **1** from *H. erinaceum*.

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Figure 2. Typical 2D NMR correlations in **1**, **1a**, and **1b**.

Hericirine (**1**) was obtained as a yellowish gum ($[\alpha]_D^{25}$: –12.9 (*c* 0.2, MeOH)).[7] The molecular structure of **1** was inferred to be $C_{29}H_{39}NO$ from a pseudomolecular ion peak $[M+H]^+$ at *m/z* 418.3107 in the HRESIMS spectrum(calcd for $C_{29}H_{40}NO$ 418.3104). The ^{1}H NMR spectrum of **1** (Table 1) revealed five aromatic protons [δ_H 7.15 (H-4', 8'), 7.20 (H-6'), 7.27 (H-5', 7')], four olefinic protons [δ_H 5.17 (H-16), 5.18 (H-11), 5.25 (H-17), 5.66 (H-3)], five methine groups [δ_H 1.44 (H-8), 1.48 (H-19), 1.88 (H-18), 2.06 (H-14), 2.54 (H-5)], five methylene groups [δ_H 1.39 (H-7a), 1.48 (H-6a), 1.87 (H-7b), 1.88 (H-6b), 2.20 (H-10a), 2.51 (H-10b), 2.86 (H-2'), 3.75 (H-1')], and five methyl groups [δ_H 0.62 (H-13), 0.83 (H-20, 21), 0.92 (H-22), 1.03 (H-15)]. The ^{13}C NMR spectrum (Table 1), obtained using the DEPT spectrum, showed 29 signals including a carbonyl group [δ_C 170.7 (C-2)], six benzene ring signals [δ_C 126.4 (C-6'), 128.5 (C-5', 7'), 128.9 (C-4', 8'), 139.0 (C-3')], three double bonds [δ_C 107.1 (C-11), 114.9 (C-3), 132.6 (C-17), 135.0 (C-16), 140.1 (C-12), 150.7 (C-4)], a quaternary carbon [δ_C 47.0 (C-9)], five methine carbons [δ_C 33.1 (C-19), 40.1 (C-14), 42.8 (C-18), 48.7 (C-5), 55.0 (C-8)], five methylene carbons [δ_C 22.3 (C-6), 28.8 (C-7), 35.2 (C-2'), 40.2 (C-10), 41.0 (C-1')], and five methyl carbons [δ_C 12.2 (C-13), 17.6 (C-22), 19.7 (C-21), 20.0 (C-20), 20.8 (C-15)]. Analyses of the ^{1}H and ^{13}C NMR spectral data indicated that **1** had characteristics of a γ–lactam (C-2, 3, 4, 12) and phenyl group (C-3'–8'), which was supported by the IR spectrum [γ–lactam (1710 cm^{-1}) and phenyl group (1595 cm^{-1})].[8] Compound **1** was similar to hericerin, an isoindolinone alkaloid from *H. erinaceum*.[9,10] However, **1** was lacking a benzene ring and side-chain compared with hericerin. In contrast, other signals of **1** were similar to the C and D rings and a side-chain of ergosterol (C-5–11 and 13–22), which is a major constituent of most mushrooms.[11,12] According to related correlations in HMQC, HMBC, COSY, and NOESY spectra, the framework of **1** was combined with hericerin and ergosterol. In the alkaloid component, HMBC correlations between H-3 (δ_H 5.66)/C-12 (δ_C 140.1), C-4 (δ_C 150.7), and C-2 (δ_C 170.7); H-1' (δ_H 3.75)/C-2' (δ_C 35.2), C-12 (δ_C 140.1), and C-2 (δ_C 170.7); and H-2' (δ_H 2.86)/C-1' (δ_C 41.0), C-4', 8' (δ_C 128.9), and C-3' (δ_C 139.0) indicated that the γ–lactam contained a double bond, and two methylene carbons linked to γ–lactam and phenyl groups. The alkaloid part was similar to the B and C rings of hericerin (**1a**), excluding a double bond located at C-3 and 4 (Fig. 2).[10] However, the sterol component showed HMBC correlations between H-13 (δ_H 0.62)/C-10 (δ_C 40.2), C-9 (δ_C 47.0), C-5 (δ_C 48.7), and C-8 (δ_C 55.0); H-6 (δ_H 1.48, 1.88)/C-9 (δ_C 47.0) and C-8 (δ_C 55.0); H-7 (δ_H 1.39, 1.87)/C-9 (δ_C 47.0) and C-5 (δ_C 48.7); and H-10 (δ_H 2.20, 2.51)/C-5 (δ_C 48.7), which was assigned to the C and D rings of sterol. According to the COSY and HMBC correlations, as well as comparisons between the chemical shifts of ^{1}H and ^{13}C NMR data, a side-chain was also identified (C-14–22). The above signals were extremely similar to the C and D rings and a side-chain of ergosterol (**1b**) (Fig. 2).[12,13] The relative stereochemistry of the sterol component was also same as ergosterol, which was deduced from NOESY experiments (Fig. 3). The HMBC spectrum revealed key correlations between two parts: H-5 (δ_H 2.54)/C-3 (δ_C 114.9), C-4 (δ_C 150.7); H-11 (δ_H 5.18)/C-12 (δ_C 140.1), and C-4 (δ_C 150.7); H-3 (δ_H 5.66)/C-5 (δ_C 48.7), C-12 (δ_C 140.1), and C-4 (δ_C 150.7); and H-10 (δ_H 2.20, 2.51)/C-12 (δ_C 140.1), and indicated that the sterol component was located at C-4, 12 of the alkaloid component (Fig. 2). Thus, the overall framework of compound **1** was determined and named hericirine. It is the first isolation of ergosterol conjunction-type alkaloid.

Figure 3. Observed key NOEs for **1** and **1b**.

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Table 1. ^{1}H and ^{13}C NMR spectral data for compound **1** in $CDCl_3$

Position	δ_H mult (*J*, Hz)[a]	δ_C[b]
2		170.7
3	5.66 s	114.9
4		150.7
5	2.54 m[c]	48.7
6	1.48 m[c]	22.3
	1.88 m[c]	
7	1.39 m[c]	28.8
	1.87 m[c]	
8	1.44 m[c]	55.0
9		47.0
10	2.20 dd (17.2, 1.0)	40.2
	2.51 dd (17.2, 6.3)	
11	5.18 d (6.3)	107.1
12		140.1
13	0.62 s	12.2
14	2.06 m	40.1
15	1.03 d (6.8)	20.8
16	5.17 dd (15.3, 7.7)	135.0
17	5.25 dd (15.3, 7.0)	132.6
18	1.88 m[c]	42.8
19	1.48 m[c]	33.1
20	0.83 d (6.5)	20.0
21	0.83 d (6.5)	19.7
22	0.92 d (6.8)	17.6
1'	3.75 t (7.5)	41.0
2'	2.86 t (7.5)	35.2
3'		139.0
4'	7.15 dd (7.2, 1.8)	128.9
5'	7.27 t (7.2)	128.5
6'	7.20 t (7.2)	126.4
7'	7.27 t (7.2)	128.5
8'	7.15 dd (7.2, 1.8)	128.9

[a] Assignment by ^{1}H NMR experiments performed at 600 MHz.

[b] Assignment by ^{13}C NMR experiments performed at 150 MHz.

[c] The coupling constants cannot be assigned due to signal overlap.

Compound **1** was examined for its inhibitory effect on NO, PGE_2, TNF-α, and IL-1β production in LPS-induced RAW264.7 cells using the Griess method and commercially available kits.[14] Compound **1** decreased NO, PGE_2, TNF-α, and IL-1β production in a dose-dependent manner based on an enzyme immunoassay in LPS-induced RAW264.7 cells (Fig. 4 and 5). L-NAME (L-NG-Nitro Arginine methyl ester), a standard NO inhibitor, was used as a positive control and caused a significant inhibition of NO production (Fig. 4A) at a concentration of 50 μM, and NS-398 (N-[2-(cyclohexyloxy)-4-nitrophenyl]-methanesulfonamide), a PGE_2 selective inhibitor, was used as a drug control and significantly inhibited the production of PGE_2 (Fig. 4B) at a concentration of 5 μM.[15] In addition, LPS treatment significantly increased iNOS and COX-2 protein expression levels, but pre-treatment with compound **1** inhibited iNOS and COX-2 protein expression (Fig. 6) based on Western blot analysis.[14] Cell viability was evaluated by determining mitochondrial reductase function with an assay based on the reduction of the tetrazolium salt 3-[4,5-dimethylthiazol-2-yl]-2,5-diphenyltetrazolium bromide (MTT) to formazan crystals. The formation of formazan from MTT is proportional to the number of functional mitochondria in living cells. The results showed that compound **1** did not exhibit significant cytotoxicity in RAW264.7 cells at the tested concentrations (data not shown). Recent studies on ethanol extract from *H. erinaceum* demonstrated that it inhibits macrophage function by decreasing the production of inflammatory mediators such as NO, prostaglandins E2 (PGE_2) and reactive oxygen species (ROS).[16] However, there were no reports concerning the active compounds on anti-inflammatory activity. Steroids show significant anti-inflammatory activity in other natural resources, including ergosterol.[17] Therefore, the significant inhibitory activity of compound **1** may be due to the presence of ergosterol.

Figure 4. The effects of **1** on nitrite (A) and PGE_2 (B) levels in RAW264.7 cells. Cells were pre-treated for 3 h with the indicated concentrations of 1 and then stimulated for 18 h with LPS (1 μg/mL). Data represent the mean ± S.D. of 3 experiments. $^{*}p < 0.05$ compared with the group treated with LPS. L-NAME and NS-398 were used as the positive controls.

Figure 5. The effects of **1** on TNF-α (A), IL-6 (B) and IL-1β (C) levels in RAW264.7 cells. Cells were pre-treated for 3 h with the indicated concentrations of 1 and then stimulated for 18 h with LPS (1 μg/mL). Data represent the mean ± S.D. of 3 experiments. $^{*}p < 0.05$ compared with the group treated with LPS.

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Figure 6. The effects of **1** on iNOS (A) expression and COX-2 (B) expression in RAW264.7 cells. Cells were pre-treated for 3 h with the indicated concentrations of **1** and then stimulated for 18 h with LPS (1 μg/mL). Data represent the mean ± S.D. of 3 experiments. $^{*}p < 0.05$ compared with the group treated with LPS.

In conclusion, a novel ergosterol conjunction-type alkaloid, hericirine (**1**), was isolated from the dried fruiting bodies of *H. erinaceum* and its structure was elucidated. Moreover, **1** exhibited significant inhibitory activity against LPS-induced expression of iNOS and COX-2, and reduced the levels of pro-inflammatory mediators and cytokines, NO, PGE2, TNF-α, IL-6 and IL-1β in RAW264.7 cells. These results may provide a scientific basis for the complement of the chemical constituents of the mushroom *H. erinaceum*, as well as for the development of novel anti-inflammatory agents.

Acknowledgments

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Supplementary Material

^{1}H-, ^{13}C-NMR, HMQC, HMBC, COSY, and HR-ESI-MS spectra of compound **1** are available as Supplementary Information.

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7. *Hericirine (**1**)*: yellowish gum; mp 122-128 °C; $[\alpha]_D^{25}$: –12.9 (*c* 0.2, MeOH); IR (KBr): ν_{max} 1710, 1595 cm^{-1}; HRESIMS *m/z* 440.2924 $[M+Na]^+$ (calcd for $C_{29}H_{39}NONa$, 440.2924); ^{1}H NMR (600 MHz, $CDCl_3$) and ^{13}C NMR (150 MHz, $CDCl_3$) spectroscopic data: see Table 1.
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9. *Hericerin (**1a**)*: pale yellow solid; mp 69−71 °C; ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.60 (s, 1H, OH), 7.30−7.15 (m, 5H, Ph), 6.41 (s, 1H, H4), 5.20 (br t, 1H, *J* = 7.2 Hz, H2′), 5.06 (br t, 1H, *J* = 7.2 Hz, H6′), 4.12 (s, 2H, H3), 3.80 (s, 3H, OMe), 3.75 (t, 2H, *J* = 7.6 Hz, H1″), 3.34 (d, 2H, *J* = 6.8 Hz, H1′), 2.99 (t, 2H, *J* = 7.6 Hz, H2″), 2.12 (m, 2H, H5′), 1.95 (br t, 2H, *J* = 7.6 Hz, H4′), 1.79 (s, 3H, H9′), 1.63 (br s, 3H, H8′), 1.55 (s, 3H, H10′); ^{13}C NMR (100 MHz, $CDCl_3$): δ 170.5 (C1), 162.4 (C5), 154.0 (C7), 140.1 (C3a), 138.9 (C3″), 135.5 (C3′), 131.4 (C7′), 128.8 (C4″, C5″), 126.7 (C6″), 124.6 (C6′), 122.0 (C2′), 116.0 (C6), 110.8 (C7a), 97.1 (C4), 56.0 (OMe), 51.4 (C3), 43.7 (C1″), 39.9 (C4′), 35.3 (C2″), 26.9 (C5′), 25.8 (C8′), 21.7 (C1′), 17.8 (C10′), 16.2 (C9′).
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11. *Ergosterol (**1b**)*: white powder; mp 153−155 °C; ^{1}H NMR (400 MHz, $CDCl_3$): δ3.64 (1H, m, H-3), 5.57 (1H, m, H-6), 5.38 (1H, m, H-7), 0.63 (3H, s, H-18), 0.95 (3H, s, H-19), 1.04 (3H, d, *J* = 6.6 Hz, H-21), 5.17 (1H, m, H-22), 5.23 (1H, m, H-23), 1.87 (1H, m, H-24), 1.48 (1H, m, H-25), 0.82 (3H, d, *J* = 6.5 Hz, H-26), 0.80 (3H, d, *J* = 6.5 Hz, H-27), 0.90 (3H, d, *J* = 6.8 Hz, H-28); ^{13}C NMR (100 MHz, $CDCl_3$) δ: 39.1 (C-1), 32.0 (C-2), 70.5 (C-3), 40.8 (C-4), 139.8 (C-5), 119.6 (C-6), 116.3 (C-7), 141.4 (C-8), 46.3 (C-9), 37.0 (C-10), 21.3 (C-11), 38.4 (C-12), 42.8 (C-13), 54.6 (C-14), 23.0 (C-15), 28.3 (C-16), 55.8 (C-17), 12.0 (C-18), 16.3 (C-19), 40.4 (C-20), 21.1 (C-21), 131.9 (C-22), 135.6 (C-23), 42.8 (C-24), 33.1 (C-25), 19.6 (C-26), 19.9 (C-27), 17.6 (C-28).
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Figure 1. Structure of compound **1** from *H. erinaceum*.

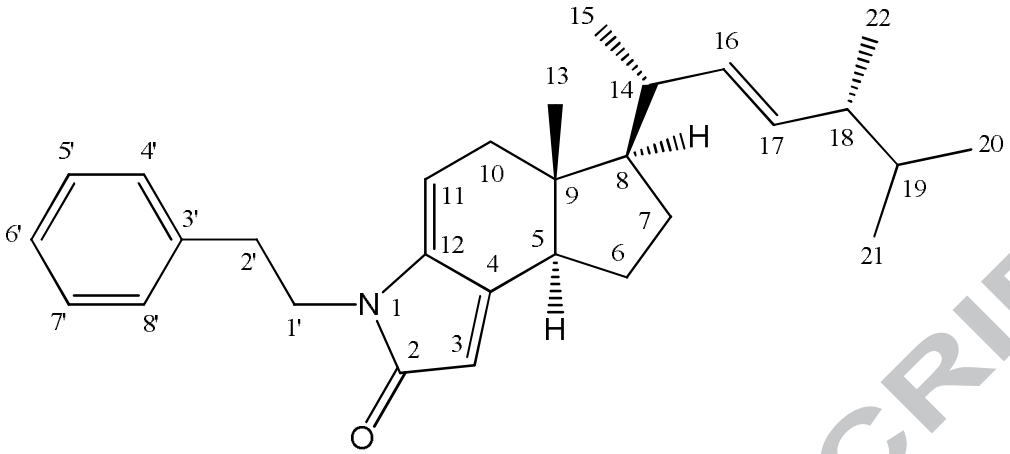

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Figure 2. Typical 2D NMR correlations in **1**, **1a**, and **1b**.

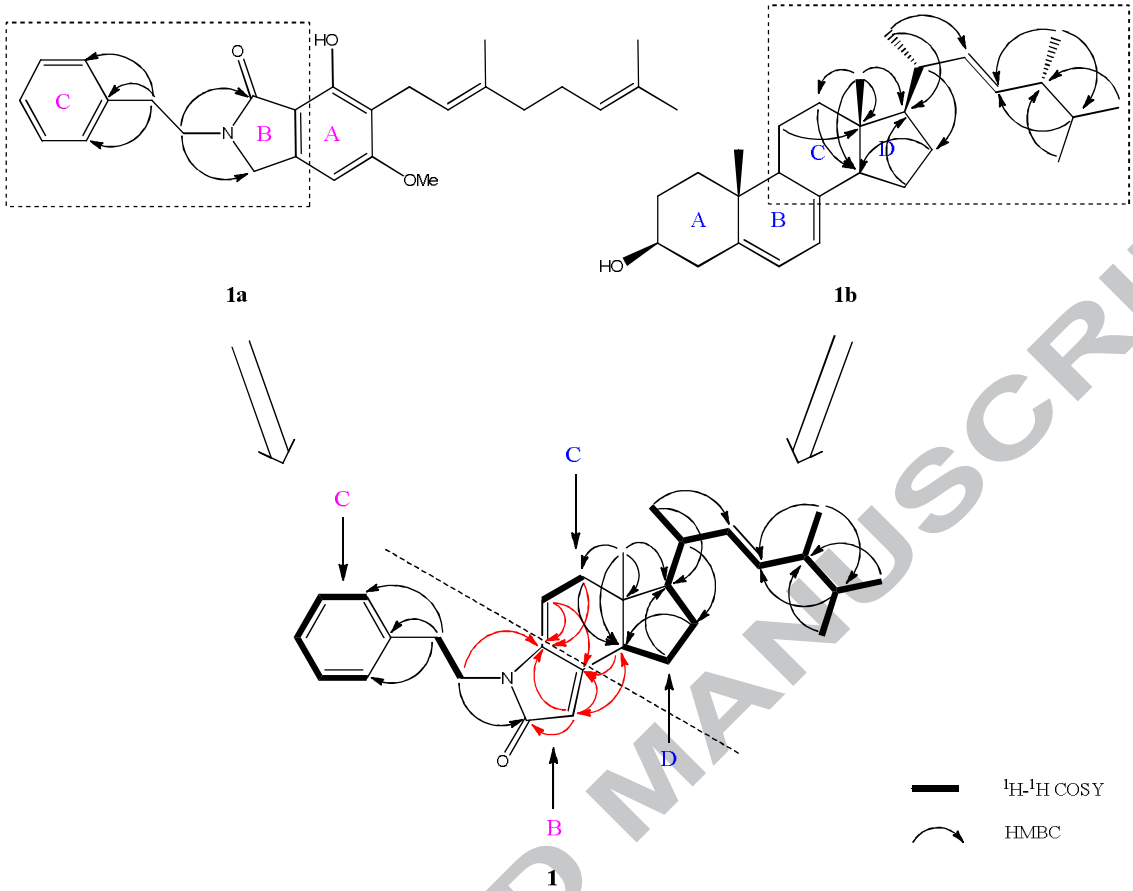

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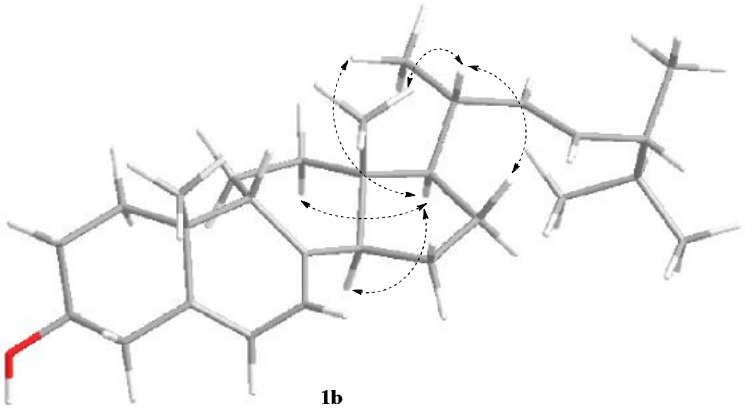
Figure 3. Observed key NOEs for **1** and **1b**.



1b

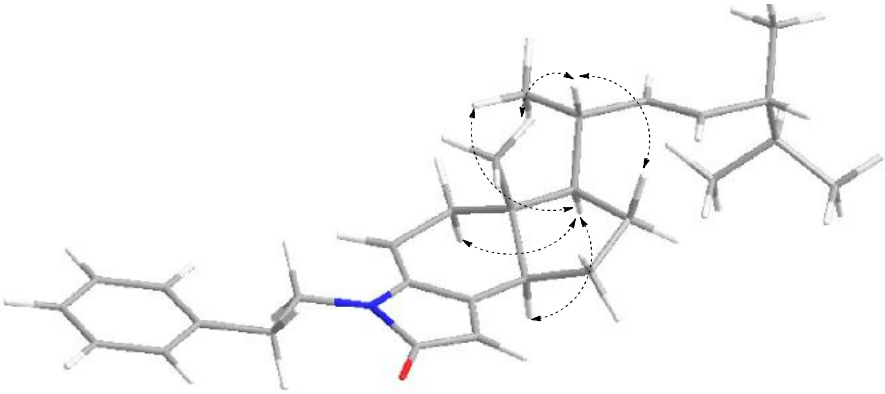


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Figure 4. The effects of **1** on nitrite (A) and PGE_2 (B) levels in RAW264.7 cells.

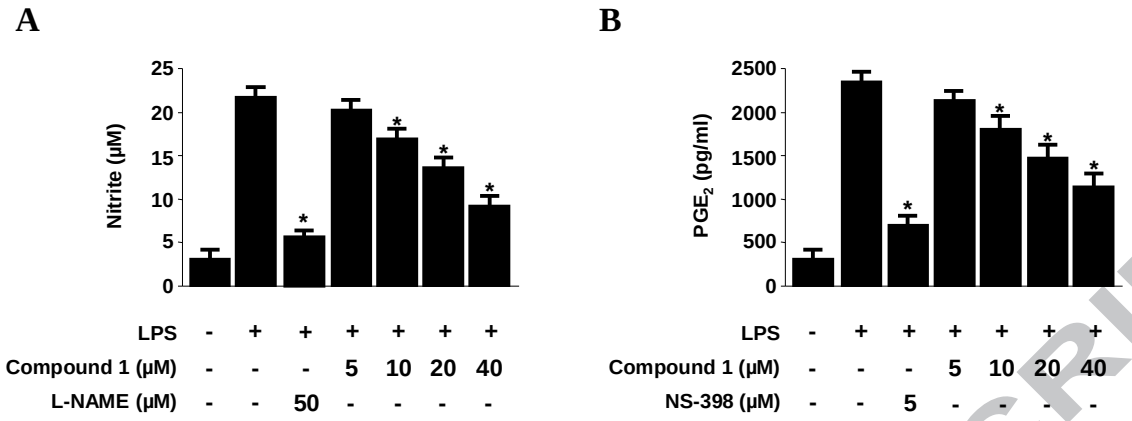

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Figure 5. The effects of **1** on TNF-α (A), IL-6 (B) and IL-1β (C) levels in RAW264.7 cells.

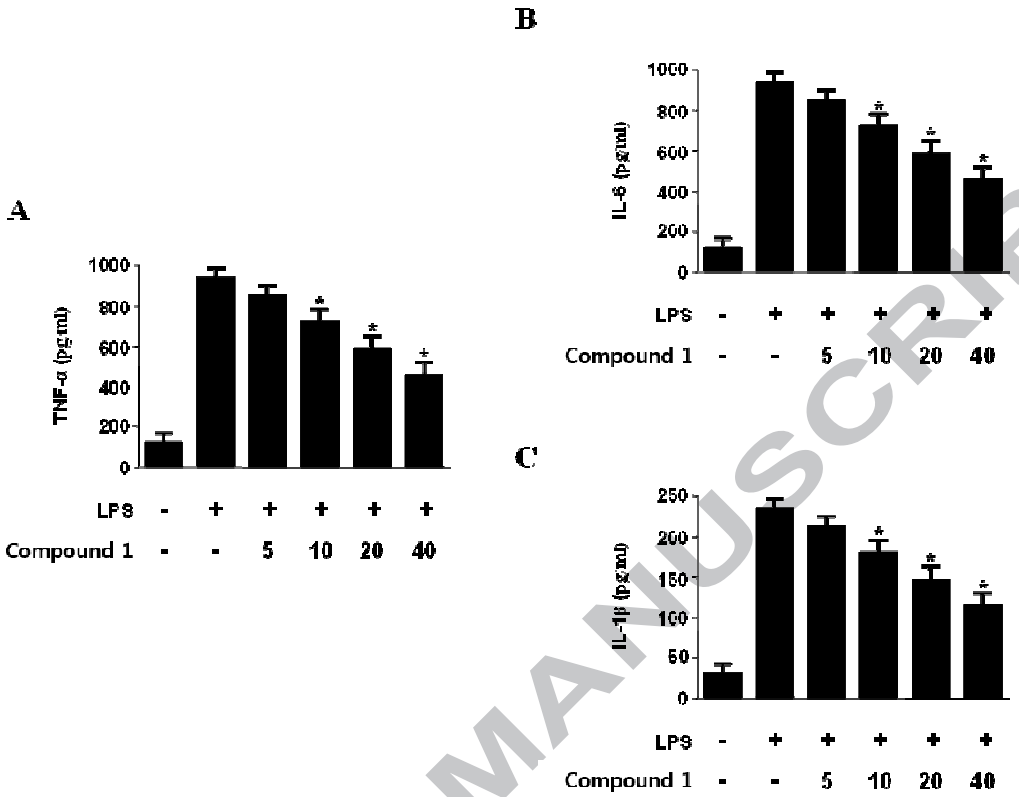

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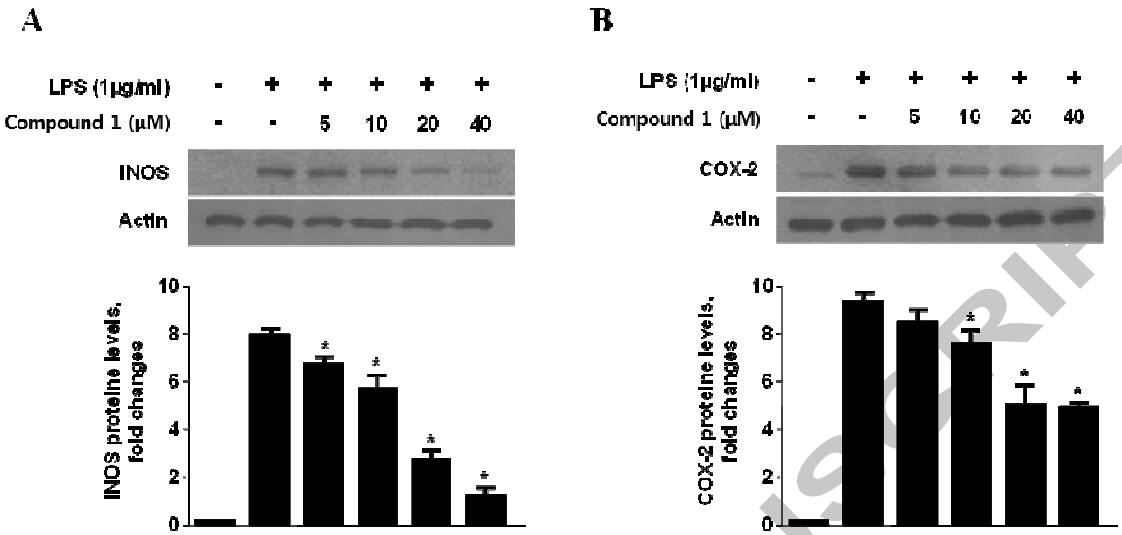
Figure 6. The effects of **1** on iNOS (A) expression and COX-2 (B) expression in RAW264.7 cells.



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